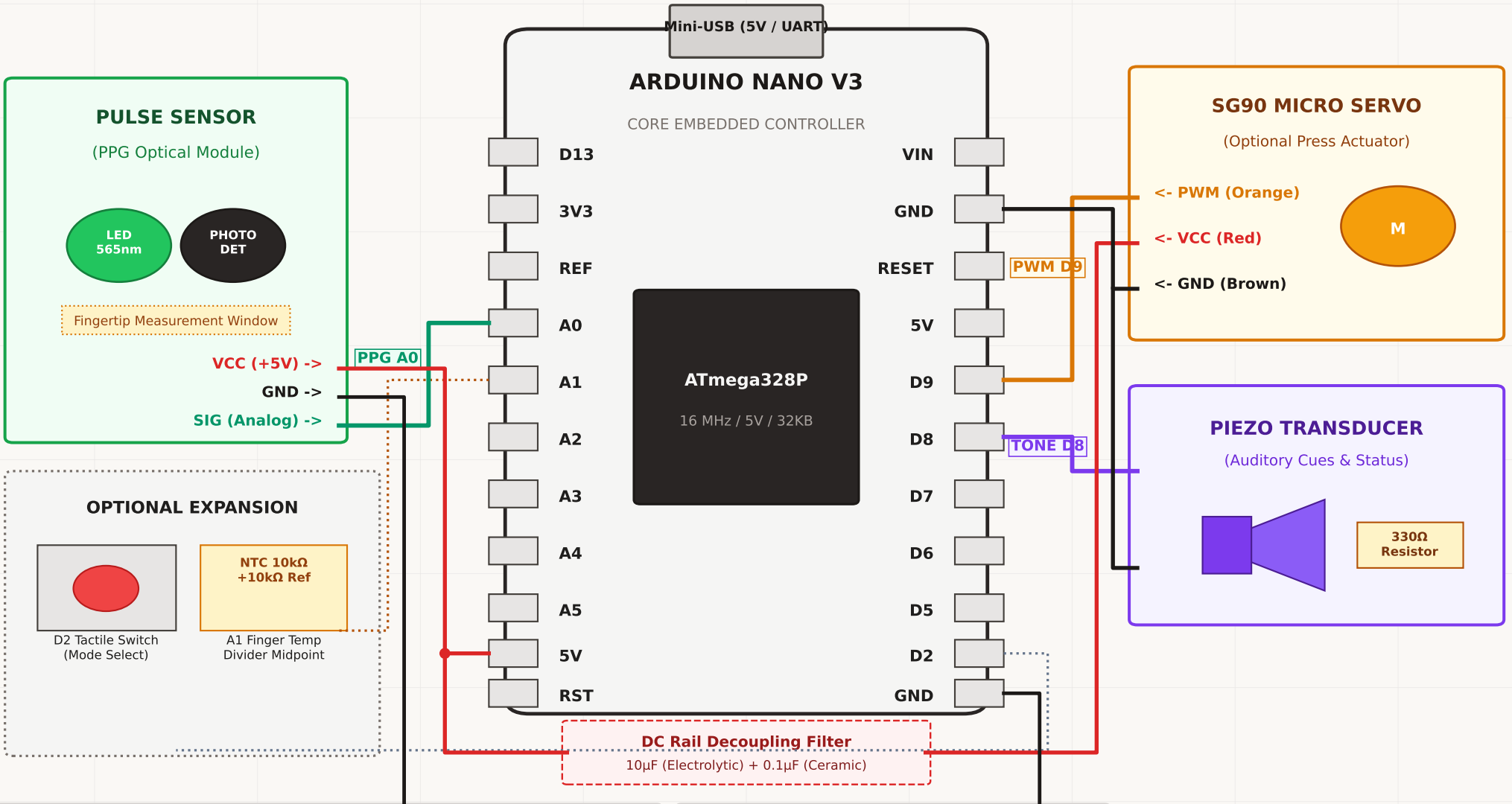
CRT-Nano: VIRTUAL SYSTEM CIRCUIT SCHEMATIC

Reflectance Photoplethysmography (PPG) Capillary Refill Time Measurement System | 23MAT204 Lab Exam



PIN ASSIGNMENT & SUBSYSTEM ROUTING

```
Pin A0   -> Optical Sensor Analog Out (PPG Reflectance)
Pin D8   -> Piezo Acoustic Transducer (+) via 330Ω Limiter
Pin D9   -> SG90 Servo Actuator PWM (Position Command)
Pin D2   -> Mode Pushbutton (Internal Pull-Up to GND)
Pin A1   -> Optional NTC 10kΩ Thermistor (Temperature Context)
5V Rail  -> Common 5V Supply Rail (10µF + 0.1µF Decoupled)
GND Rail -> Star-Ground Reference Bus
```

BILL OF MATERIALS & COST (INR)

```
• Arduino Nano V3 (CH340)    : Rs. 155
• PPG Pulse Sensor Module    : Rs.  75
• Piezo Acoustic Disc        : Rs.  12
• Premium Jumper Wires       : Rs.  35
• Carbon Film Resistor Pack  : Rs.  10
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CORE MANUAL CONFIG           : Rs. 287
With SG90 Micro Servo (+Rs70): Rs. 357
[Target T6 <= Rs. 360: FULLY COMPLIANT]
```

AMRITA SCHOOL OF AI

23MAT204 - Healthcare AI / Edge Challenge

```
PROJECT : CRT-Nano (Capillary Refill)
STUDENT 1: M. Dinesh Kumar (25126)
STUDENT 2: M. V. Sai Sujan (25129)
INSTRUCTOR: Dr. Suman Dutta
REVISION : v2.4 (Virtual Schematic)
DATE     : October 30, 2026
SHEET    : 1 OF 1 (Single Page PDF)
```